

Object Oriented Programming

CSIT — 2nd Semester — 2081 — Answer Sheet
Subject Code: CSC166

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Define class and objects. Show the use of default, parameterized and copy constructor with your own example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Differentiate between exception and error. What is the use of virtual function? Create a class named Shape with member function draw(), then derived the class Rectangle, Circle and Square from Shape class, and override the draw() functions in all the derived classes with appropriate messages.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. How do you convert basic data type to another basic data type? Create a class named Point with two data members x and y. Overload the binary addition operator to add the two objects of Point class.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. How do you open, close, read and write in file?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Use the concept of function template to find the difference of any two integers.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Show the use of static data members with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Can we use structure instead of class? What might be the limitations of object oriented programming approach? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Illustrate with an example to show the concept of multi-level inheritance.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Define namespace and its uses. What are the uses of new and delete?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Write a program to create a text file and write the string given by user.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Discuss the three types of access specifiers in inheritance.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. What is data conversion? When do we use inline function? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.